

Low-Discrepancy Sampling on Higher-Dimensional Spheres

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Section 1

Motivation

Low-Discrepancy Sequences

- A **deterministic** sequence of points that fills a space more uniformly than random sampling
- Three desirable properties:
 - **Uniformity** – points are spread evenly, minimising clusters and gaps
 - **Determinism** – the sequence is reproducible from a fixed starting point
 - **Incrementality** (*the key property*) – points can be appended while the whole set stays evenly distributed
- Drawback: generation becomes slower as the number of points grows

Applications of LDS on S^n

- **Robot motion planning** on S^3 and $SO(3)$ [1] – for path planning and attitude control
- **Wireless coding** – spherical codes and MIMO codebooks [2]
- **Multivariate empirical mode decomposition** [3] – direction vectors on higher-dimensional spheres via the **cylindrical mapping**
- **Filter bank design** [4] – the same cylindrical mapping underlies its filter-bank analysis
- **Statistical and machine learning** – deterministic coverage of a normalised parameter space

The Challenge in Higher Dimensions

- Sampling the 3-D sphere ($n = 2$) is well understood
- Little is known for $n > 2$ – the *curse of dimensionality*
- The **cylindrical mapping does not generalise** to higher dimensions
- Goal: one construction on S^n that is uniform, deterministic and incremental

Section 2

Overview of Low-Discrepancy Sequences

The van der Corput Sequence

- The one-dimensional low-discrepancy sequence on $[0, 1]$
- Constructed by reversing the base- b digits of the integers, with b usually prime
- Introduced by the Dutch mathematician **Johannes van der Corput** (1935)



Unit Circle S^1

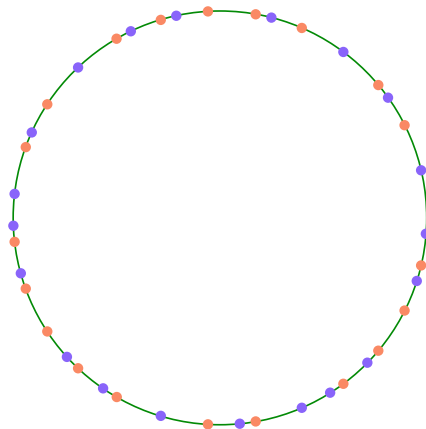
- Treat the vdC value as an angle:

$$\theta = 2\pi \cdot \text{vdc}(k, b)$$



$$[x, y] = [\cos \theta, \sin \theta]$$

- Incremental and uniform by construction

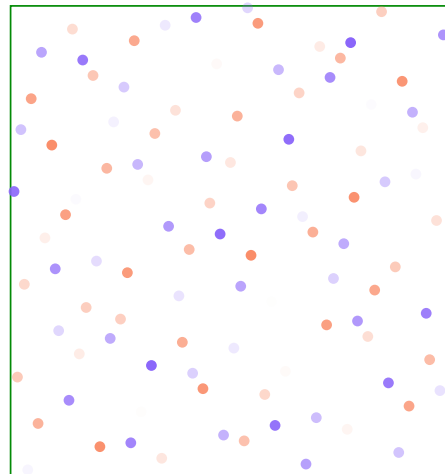


Halton Sequence on $[0, 1]^n$

- Combine n van der Corput sequences with **coprime** bases:

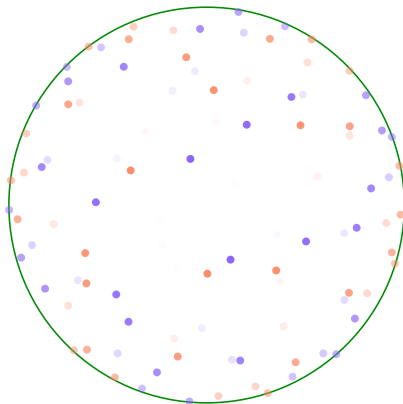
$$[x_1, \dots, x_n] = [\text{vdc}(k, b_1), \dots, \text{vdc}(k, b_n)]$$

- Uniform over the unit hypercube
- Basis of quasi-Monte Carlo (QMC) methods



Unit Sphere S^2 – Cylindrical Mapping

- Two vdc values give z and φ :
 - $z = 2 \text{vdc}(k, b_2) - 1 \in [-1, 1]$
 - $\varphi = 2\pi \text{vdc}(k, b_1) \in [0, 2\pi)$
- Radius of the horizontal circle:
 $r = \sqrt{1 - z^2}$
- $$[x, y, z] = [r \cos \varphi, r \sin \varphi, z]$$
- Used in computer graphics [5]



S^3 and $SO(3)$ – Hopf Fibration

- Hopf coordinates [1]:
 - $x_1 = \cos(\theta/2) \cos(\psi/2)$
 - $x_2 = \cos(\theta/2) \sin(\psi/2)$
 - $x_3 = \sin(\theta/2) \cos(\varphi + \psi/2)$
 - $x_4 = \sin(\theta/2) \sin(\varphi + \psi/2)$
- S^3 is a principal circle bundle over S^2
- Works for S^3 only: the cylindrical mapping does **not** extend to S^n for $n > 2$

Section 3

Previous Work

Related Work

- **Low-discrepancy sequences & quasi-Monte Carlo**
 - van der Corput (1935) [6]; Halton (1960) [7]; Sobol' (1967) [8]
- **Equidistribution & designs on the sphere**
 - Cui & Freeden [9]; Brauchart et al. [10]; Saff & Kuijlaars [11]
- **Cylindrical & Hopf mappings**
 - Wong, Luk & Heng [5]; Mitchell [12]; Yershova et al. [1]; Shoemake [13]

Related Work (cont.)

- **Random sampling on S^n**
 - normalise a Gaussian vector; rejection method [14], [15]
- **Spherical codes & applications**
 - Grassmannian beamforming [16], [17]
 - space-time codes from spherical codes [2]
 - multivariate EMD via cylindrical-mapped LDS [3], [4]

The Gap Addressed Here

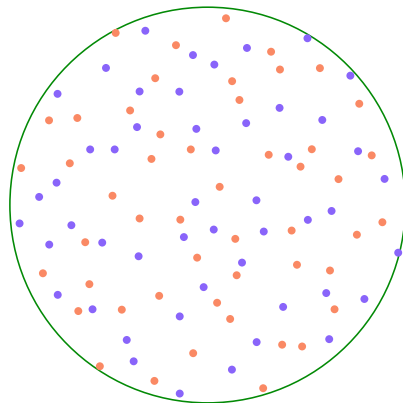
- No previous construction is **uniform + deterministic + incremental** on S^n for **arbitrary** n
- Cylindrical and Hopf mappings exploit structure special to S^2 / S^3
- Spherical designs and energy minimisation are **not incremental** (adding a point changes the configuration)
- Random sampling is **not deterministic**
- This work: van der Corput + **recursive tabulated inverse CDF**

Section 4

Our Approach

Uniform Sampling on a Unit Disk

- Surface element: $dA = r dr d\theta$
- Integrating gives $\int r dr = \frac{1}{2}r^2$, so the radial weight is r
- Cancel it with the **inverse**: use
$$r = \sqrt{\text{vdc}(k, b_2)}$$
- $\theta = 2\pi \text{vdc}(k, b_1)$
- $[x, y] = [r \cos \theta, r \sin \theta]$



Why Cylindrical Mapping Works Only for S^2

- Write a point of S^n as $p = (\cos \theta_n, \sin \theta_n u)$, $u \in S^{n-1}$
- The surface element separates into a height part and a spherical part:

$$d^n A = \sin^{n-1} \theta_n d\theta_n dA_{n-1}(u)$$

- Substituting the height $z = \cos \theta_n$:

$$d^n A = (1 - z^2)^{(n-2)/2} dA_{n-1}(u) dz$$

- The height weight $(1 - z^2)^{(n-2)/2}$ is constant **only when** $n = 2$
- S^2 : $d^2 A = d\varphi dz$ – uniform z and φ is exact (the cylindrical mapping)
- $n \geq 3$: uniform height is **biased** (over-samples the poles)
 - draw θ_n by inverting the CDF of $\sin^{n-1} \theta_n$, i.e. f_{n-1}

Recursive Construction on S^n

- Hyperspherical coordinates:
 - $x_0 = \cos \theta_n$
 - $x_1 = \sin \theta_n \cos \theta_{n-1}$
 - ...
 - $x_n = \sin \theta_n \sin \theta_{n-1} \cdots \sin \theta_1$
- Surface element:

$$d^n A = \sin^{n-1} \theta_n \sin^{n-2} \theta_{n-1} \cdots \sin \theta_2 d\theta_1 \cdots d\theta_n$$

- It factorises, but the inverse has **no closed form** for $m \geq 2$
- Peel off one angle at a time: $p_n = [\cos \theta_n, \sin \theta_n \cdot p_{n-1}]$

How to Generate the Point Set

- Let $f_m(\theta) = \int_0^\theta \sin^m \varphi \, d\varphi$, defined recursively:

$$f_m(\theta) = \begin{cases} \theta & m = 0, \\ -\cos \theta & m = 1, \\ \frac{1}{m}(-\cos \theta \sin^{m-1} \theta + (m-1)f_{m-2}(\theta)) & m \geq 2. \end{cases}$$

- Angle θ_j ($j = 2, \dots, n$) carries weight $\sin^{j-1} \theta_j$: map $\text{vdc}(k, b_j)$ onto f_{j-1}
- Invert numerically: $\theta_j = f_{j-1}^{-1}(t_j)$ by **table lookup**; f_0, f_1 are closed forms
- Assemble $p_j = [\cos \theta_j, \sin \theta_j \cdot p_{j-1}]$ up to p_n

Table Lookup: Numerical Mechanics

- **Resolution:** every table is on a **300-point** grid of $[0, \pi]$
 - spacing $h = \pi/299 \approx 1.05 \times 10^{-2}$ rad
- **Interpolation:** binary search + linear interpolation
 - $O(\log 300) \approx 9$ comparisons, clamped at the endpoints
- **Precision:** piecewise-linear error $O(h^2)$
 - max angular error $\approx 3.4 / 2.7 / 2.3 \times 10^{-4}$ rad for $m = 2, 3, 4$
- **Memory:** 300 doubles ≈ 2.4 kB per table
 - $O(n)$ tables, under 20 kB for $n \leq 5$; built lazily and cached

Implementation

- Van der Corput generator – uniform values in $[0, 1]$
- Interpolation routines – invert the cached tables
- SphereGen – abstract interface (pop, reseed)
- Sphere3 – explicit generator for S^3 ; SphereN – recursive chain bottoming out at Sphere3
- Any dimension, limited only by memory

Reference Implementations

- Four sibling projects, all implementing the same generators:
 - `lds-gen` (Python) – `locks`, `pop_batch()`
 - `lds-rs` (Rust) – `AtomicU64`, `SphereN`
 - `lds-cpp` (C++20) – `constexpr`, precomputed tables
 - `lds-gen-cpp` (C++20) – includes the `sphere_n` module
- Core generators in all four: `VdCorput`, `Halton`, `Circle`, `Disk`, `Sphere`, `Sphere3Hopf`, `HaltonN`
- The recursive `Sphere3/SphereN` exist in Python, Rust and `lds-gen-cpp` only

Cross-Language Verification and Performance

- With a fixed seed the core generators are **bit-identical** across languages
- The recursive sphere generators agree to 15+ decimal places; one coordinate differed at the 16th (machine epsilon)
- Nanoseconds per point:
 - Sphere3 (S^3): C++ 657, Rust 1035, Python 7749
 - SphereN (S^4): Rust 1607, C++ 1665, Python 12153
 - SphereN (S^5): Rust 2020, C++ 2312, Python 48443
- C++ templates win for shallow generators; Rust's lock-free counter wins for deep recursion



Section 5

Numerical Experiments

Dispersion Measure

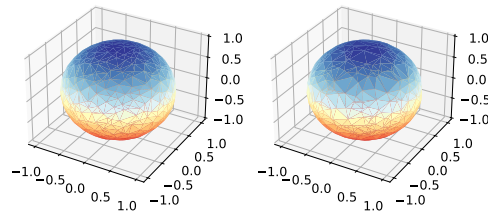
- Generate N points and build their convex hull (`scipy.spatial.ConvexHull`)
- Dispersion is the spread of neighbour distances:

$$\max_{a \in \mathcal{N}(b)} \{D(a, b)\} - \min_{a \in \mathcal{N}(b)} \{D(a, b)\}, \quad D(a, b) = \sqrt{1 - a^\top b}$$

- Lower dispersion means a more uniform point set

Random vs LDS ($N = 600$)

- **Random**: normalise a Gaussian vector (uniform on S^n)
- **LDS**: SphereN vs the CylindN baseline
- Left: ours, right: random

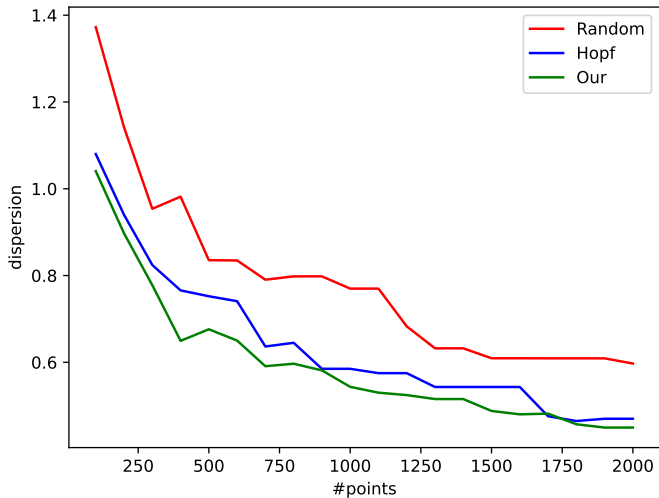


Dispersion at $N = 600$ (30 random trials)

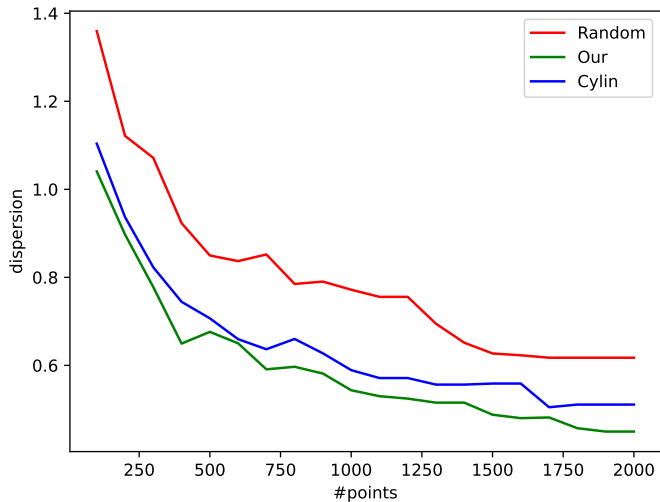
Sphere (bases)	Random	CylindN	SphereN
S^3 (2,3,5)	0.845 ± 0.041	0.659551	0.650145
S^4 (2,3,5,7)	1.090 ± 0.038	1.050584	0.912591
S^5 (2,3,5,7,11)	1.252 ± 0.041	1.358791	1.035655

- Lower is better; SphereN is lowest on every sphere
- Beats the random **mean** by more than one standard deviation

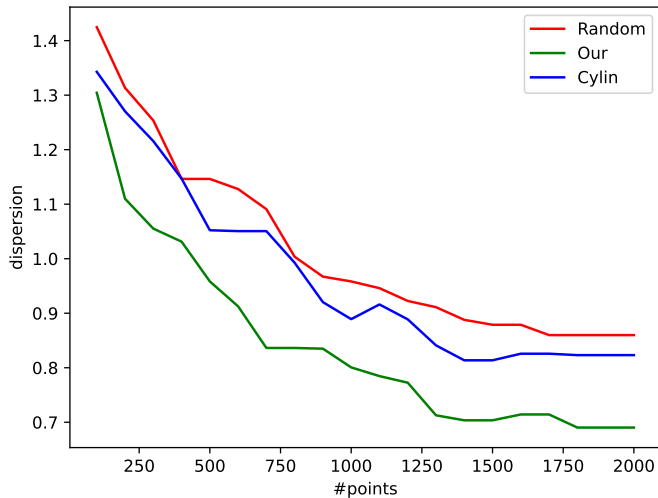
Results: S^3 vs Hopf Coordinate Method



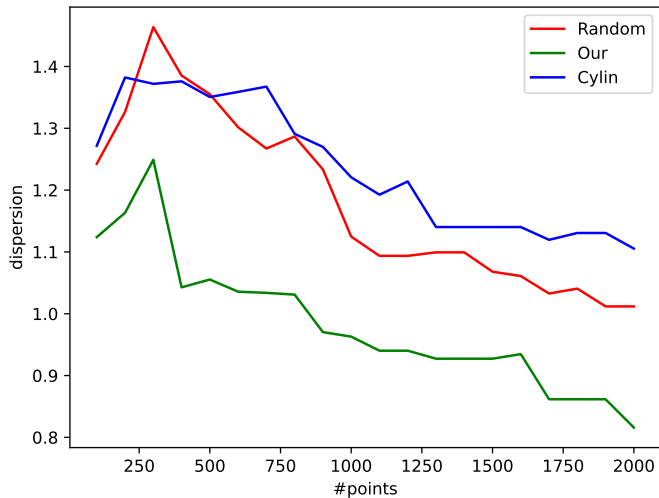
Results: S^3 vs Cylindrical Mapping



Results: S^4 vs Cylindrical Mapping



Results: S^5 vs Cylindrical Mapping



Section 6

Conclusions

Conclusions

- A uniform, deterministic and **incremental** LDS on S^n
- Built from the van der Corput sequence plus a recursive spherical construction
- Outperforms random sampling, especially when the number of points is small
- Superior to the Hopf method on S^3 and to cylindrical mapping on S^4 and S^5

Future Work

- Monte Carlo integration, optimisation and machine learning
- Faster table lookup or closed-form approximations for large n
- Extensions to $SO(n)$ and other manifolds

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